

ECON 311 - Winter 2024
Final Exam

Question I: Long-run Growth

a)

From production function: $\frac{Y_t}{\bar{L}} = y_t = A_t (1-\bar{i})$

$$g_y = g_A + \underbrace{g_{(1-\bar{i})}}_{=0}$$

$$g_y = g_A$$

To calculate g_A , we use the LOM for productivity

$$\frac{A_{t+1} - A_t}{A_t} = g_A = \bar{z} \bar{i} \bar{L} = g_y$$

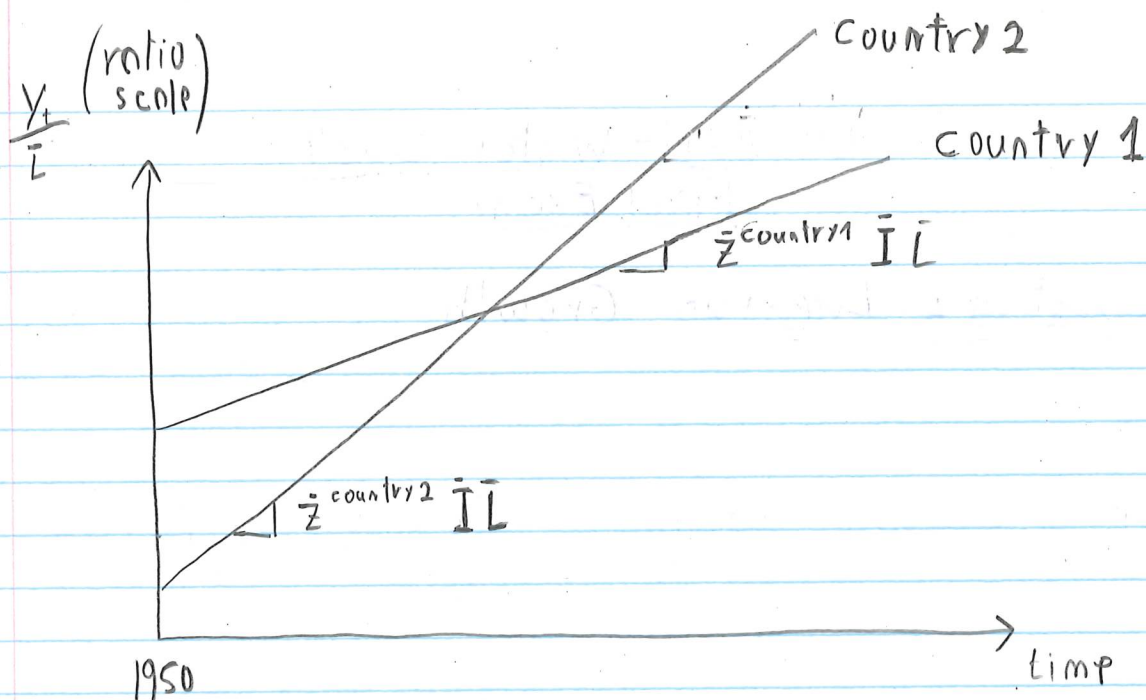
b)

$$\text{In 1950: } \frac{Y_{1950}^{\text{country 1}}}{\bar{L}} = A_{1950}^{\text{country 1}} (1-\bar{i}) > A_{1950}^{\text{country 2}} (1-\bar{i}) = \frac{Y_{1950}^{\text{country 2}}}{\bar{L}}$$

However,:

$$g_y^{\text{country 1}} = \bar{z}^{\text{country 1}} \bar{i} \bar{L} < \bar{z}^{\text{country 2}} \bar{i} \bar{L} = g_y^{\text{country 2}}$$

Therefore, country 2 will eventually have more output per capita



Question II: Cross-Country Income Differences

$$\frac{Y_B}{Y_{US}} = 0.01$$

$$\left(\frac{K_B}{K_{US}}\right)^{1/3} = 0.18$$

To solve for $\frac{\bar{A}_B}{\bar{A}_{US}}$, we use the production

function as follows:

$$\underbrace{\frac{Y_B}{Y_{US}}}_{0.01} = \frac{\bar{A}_B}{\bar{A}_{US}} \underbrace{\left(\frac{K_B}{K_{US}}\right)^{1/3}}_{0.18}$$

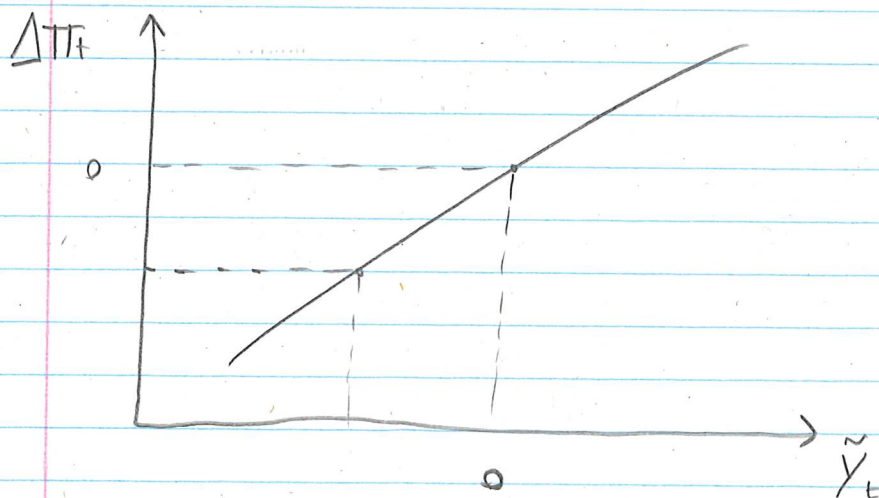
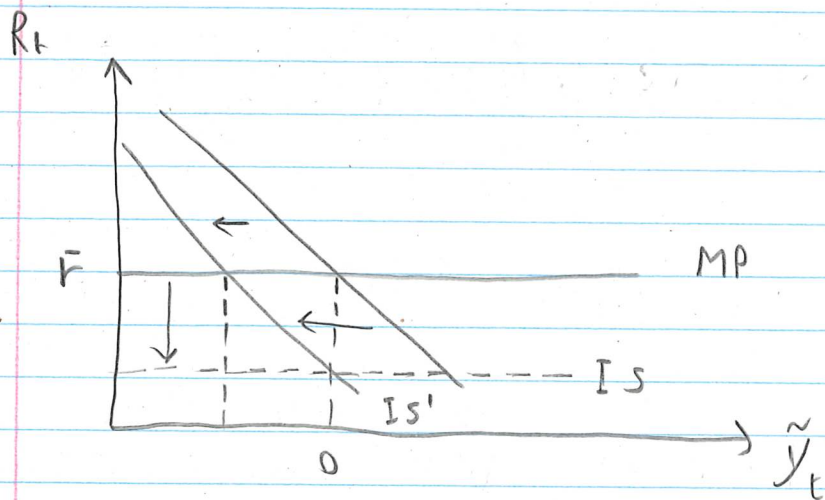
$$\frac{\bar{A}_B}{\bar{A}_{US}} = \frac{0.01}{0.18} = 0.056$$

Question III: International Transmission of Monetary Policy

$\downarrow i^{US} \Rightarrow \downarrow R^{US} \Rightarrow \frac{E_{\$/\text{US}}}{\text{US}} \downarrow$ (demand for Canadian Dollars $\uparrow$, this is an appreciation)

$\Rightarrow E_{\$/\text{US}} \frac{P_{US}}{P_{\$/\text{US}}} \downarrow$ (due to sticky prices, this means that Canadian goods are relatively more expensive than US goods)

This is equivalent to a negative demand shock



To return to $\tilde{y}_t$, the Bank of Canada must also $\downarrow i^{CAN} \Rightarrow \downarrow R_t^{CAN}$ (dotted line in graph)

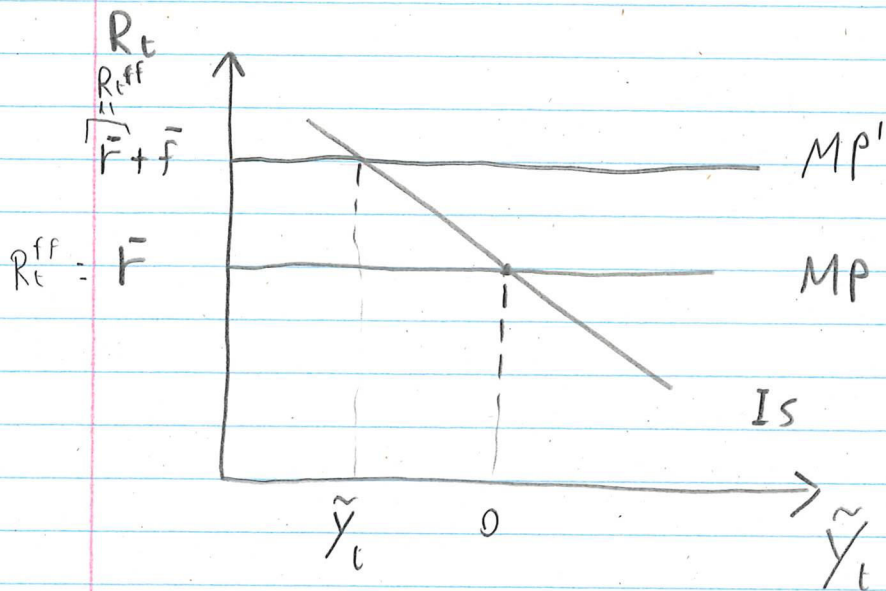
Question IV: Financial Crisis

With financial frictions, the Central Bank is only able to affect R_t^{ff} , but not the R_t , which is the real interest rate in the market.

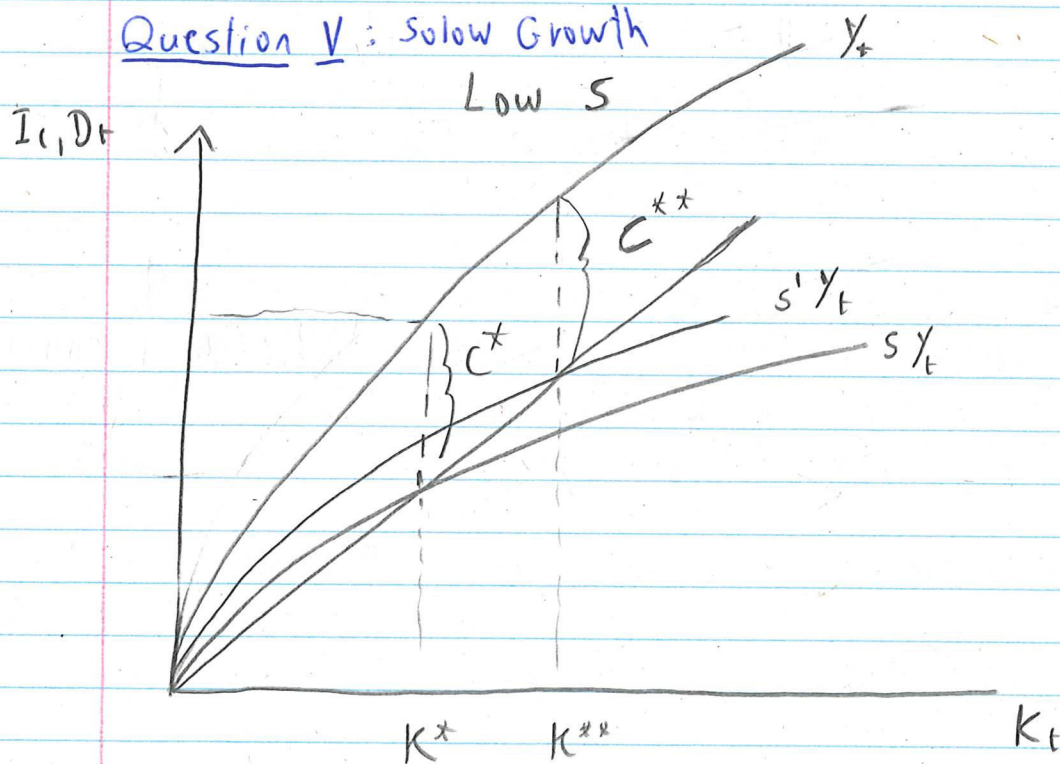
$$R_t = R_t^{ff} + \bar{f}$$

Substituting for this in the IS-Curve

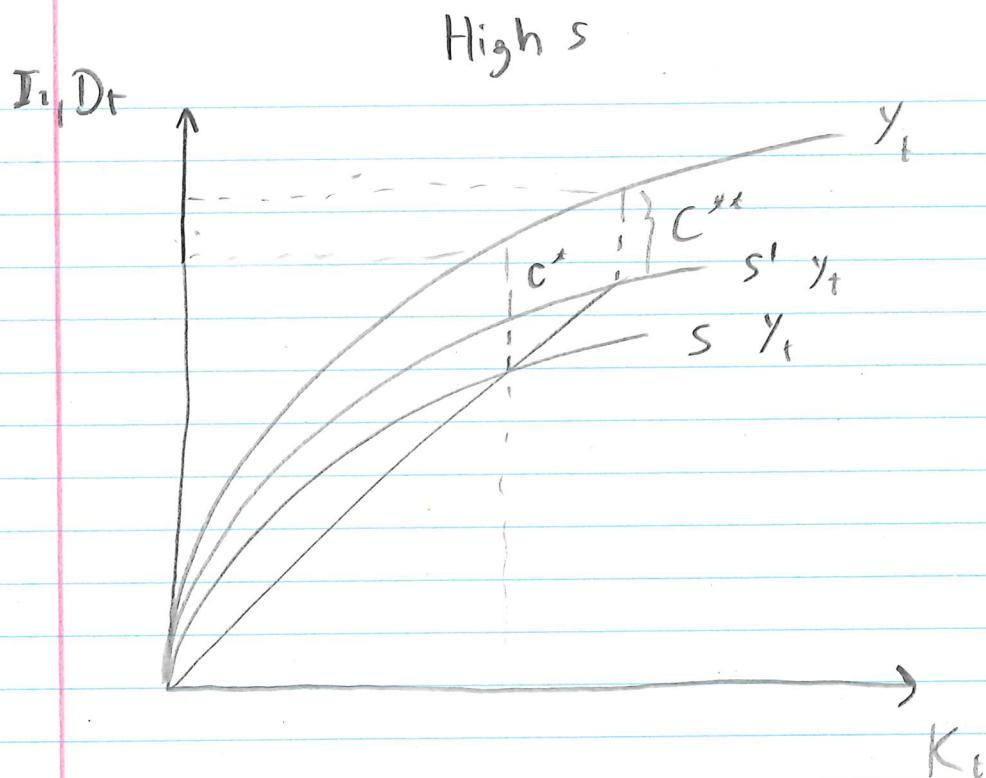
$$\tilde{y}_t = \frac{1}{1 - b_c} [\bar{a} - \bar{b}_i ((R_t^{ff} + \bar{f}) - \bar{r})] - 1$$



Question V : Solow Growth



C increases in the long run



C decreases in the long run

$\uparrow s$ - two effects

- 1) $\downarrow C$ (less share of output for consumption)
- 2) $\uparrow C$ (move capital in LR $\Rightarrow \uparrow$ output in LR)

Prime Minister is never correct as GDP does not decrease in the short run. Whether consumption increases in the long run depends on which effect is larger (1) or 2).

As shown in figures $\Rightarrow$ low $s \Rightarrow$ 2) is larger than 1)

$\Downarrow$
high $s \Rightarrow$ 1) is larger than 2)

Part II: Long-Answer Questions

Q I) IS-MP Model

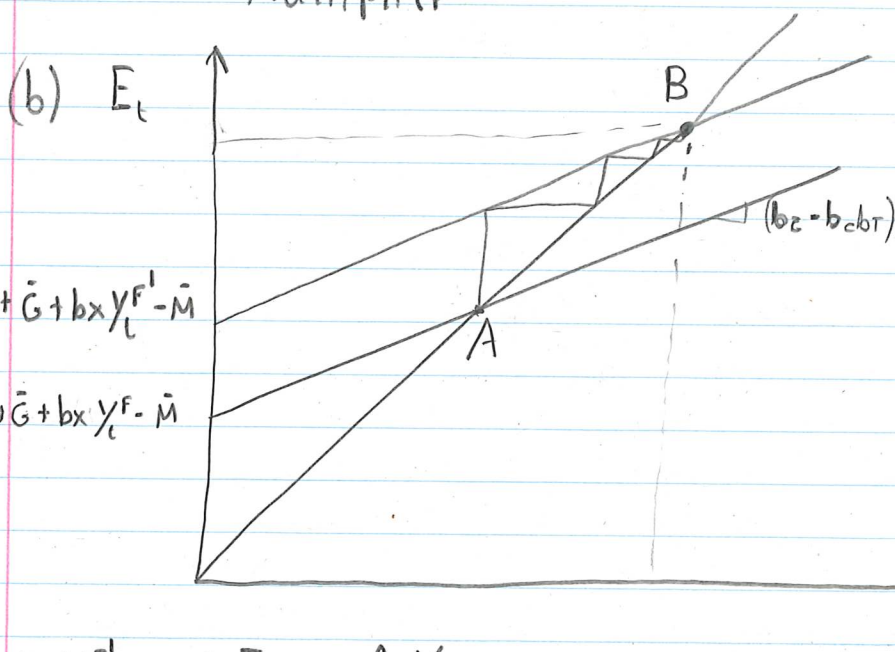
(a)

$$E_t = a_c + b_c Y_t - b_c b_r Y_t + a_i - b_i R_t + \bar{G} + b_x Y_t^F - \bar{M}$$

In equilibrium $Y_t = E_t$

$$Y_t - b_c Y_t + b_c b_r Y_t = a_c + a_i - b_i R_t + \bar{G} + b_x Y_t^F - \bar{M}$$

$$Y_t = \underbrace{\frac{1}{(1 - b_c + b_c b_r)}}_{\text{Multiplier}} [a_c + a_i - b_i R_t + \bar{G} + b_x Y_t^F - \bar{M}]$$



$\uparrow Y_t^F \Rightarrow \uparrow E_t \Rightarrow \uparrow Y_t$ this increases E_t again, but by less than the initial increase which increases Y_t again, but by less than before.

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This process continues until it reaches the new equilibrium at point B

(c)

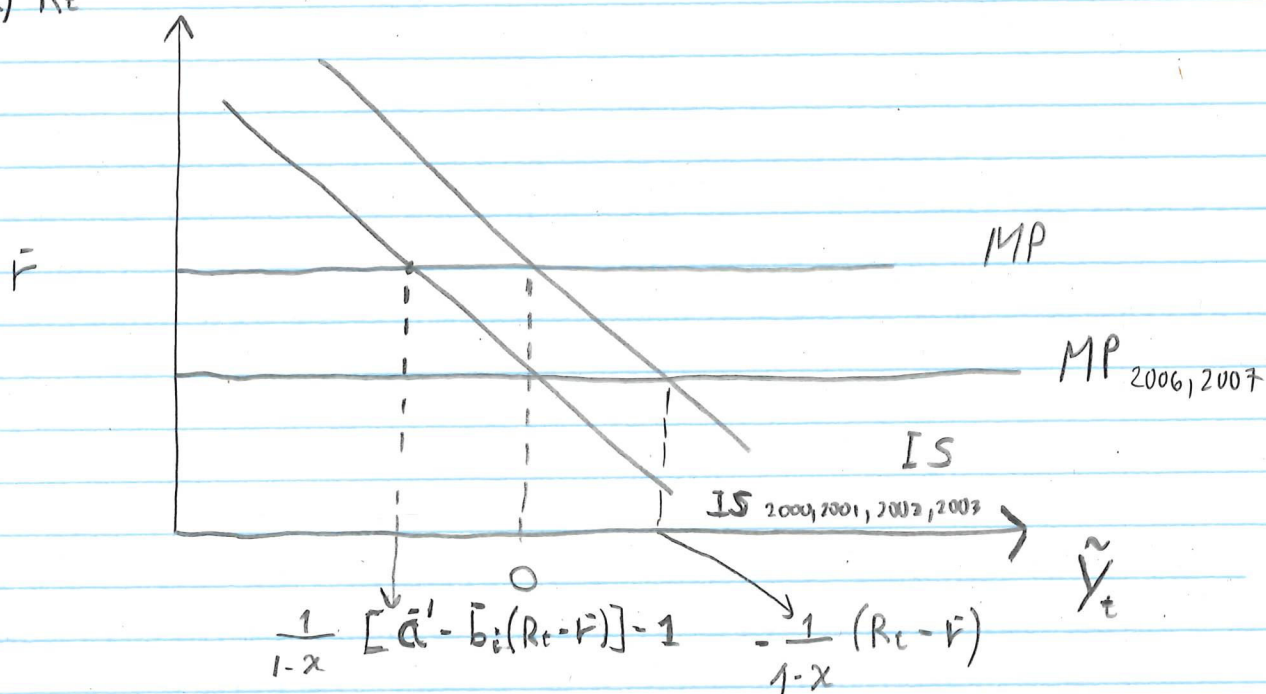
$$\frac{y_t}{\bar{y}} - 1 = \frac{1}{1-\alpha} \left[\underbrace{\frac{a_c + a_i - b_i \bar{r} + \bar{G} + b_x y_t^F - \bar{M}}{\bar{y}}}_{\bar{a}} + \underbrace{\frac{b_i (R_t - \bar{r})}{\bar{y}}}_{\bar{b}_i} \right] - 1$$

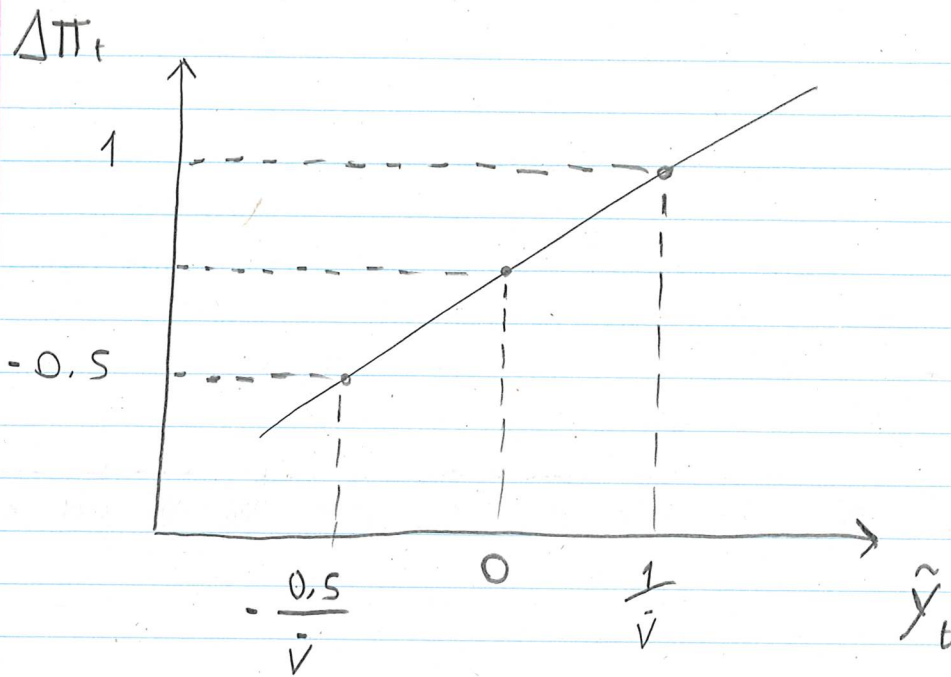
Where $\bar{y} = \frac{1}{1-\alpha} [a_c^{LR} + a_i^{LR} - b_i^{LR} \bar{r} + \bar{G}^{LR} + b_x^{LR} \bar{y}^F - \bar{M}^{LR}]$

and $\alpha = (b_c - b_c b_i)$

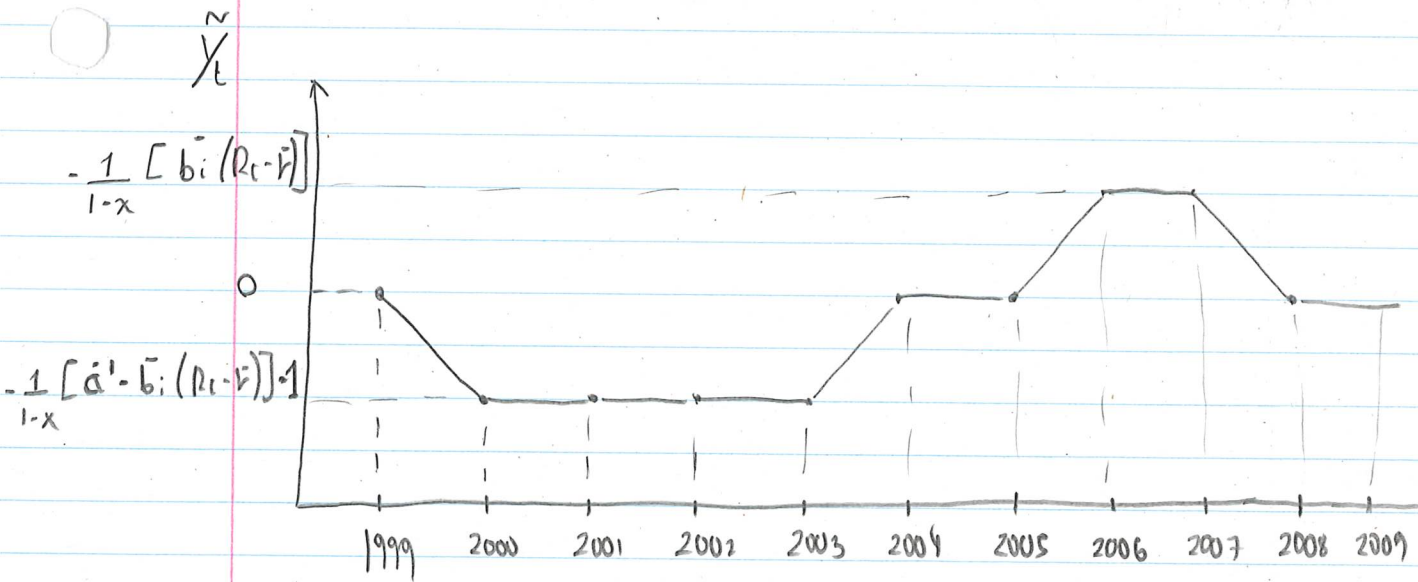
$$\Rightarrow \tilde{y}_t = \frac{1}{1-\alpha} [\bar{a} - \bar{b}_i (R_t - \bar{r})] - 1$$

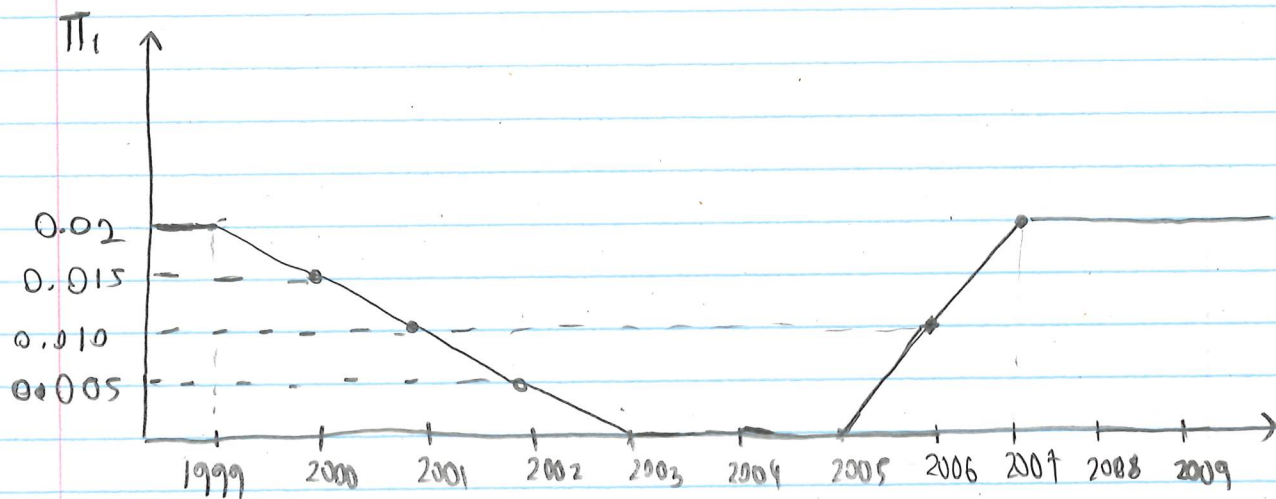
(d) R_t





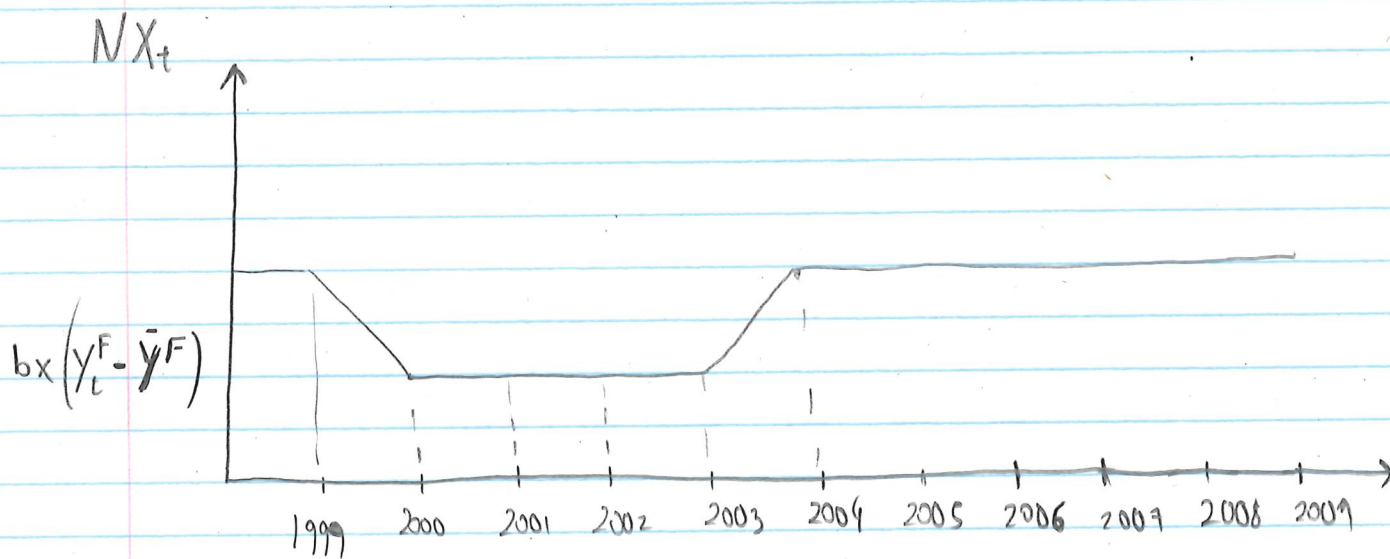
$$\Delta\pi_t = \bar{v} \tilde{y}_t$$





(e)

$$NX_t = (b_x y_t^F - b_x \bar{y}^F) + (M_t - \bar{M})$$



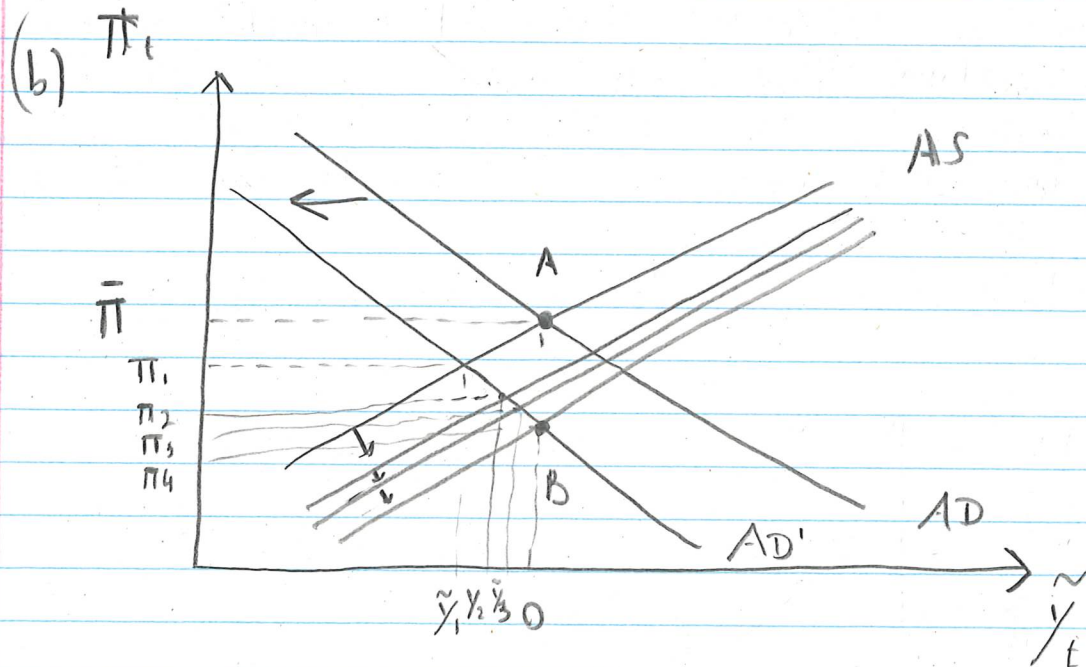
QII) AS-AD Model

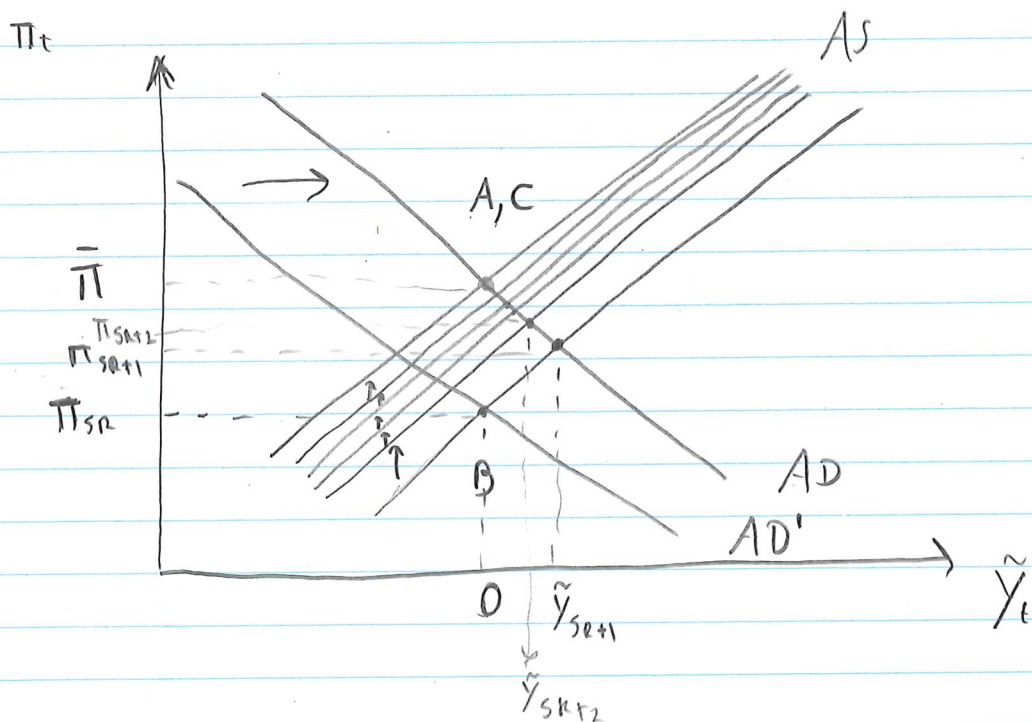
(a)

$$\text{IS Curve: } \tilde{y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i (R_t - \bar{r})] - 1$$

Substituting for the policy rule;

$$\text{AD Curve: } \tilde{y}_t = \frac{1}{1-b_c} [\bar{a} - \bar{b}_i \bar{m} (\pi_t - \bar{\pi})] - 1$$





If $\bar{m} \uparrow \Rightarrow$ flatter AD

This means that the government responds more to deviations from the inflation target. Therefore, the deviations from $\bar{\pi}$ in the previous answer would be smaller and the deviations from $\tilde{y}_t = 0$ would be larger.

(Aside $\Rightarrow$ goes back to C faster as well)

